Stablecoin Dynamics and Liquidity Fragmentation: Evidence from the March 2023 SVB Crisis

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Abstract

As the name suggests, stablecoins are intended to provide a stable, frictionless dollar-equivalent asset for digital markets. However, the March 2023 Silicon Valley Bank (SVB) crisis exposed their reliance on reserve quality and market confidence. Using high-frequency BTC spot data across major exchanges, we study the microstructure of this crisis across BTC/USD, BTC/USDT, and BTC/USDC markets. We find that liquidity fragmentation and pricing disruptions were highly concentrated in stablecoin-quoted pairs, characterized by widened cross-currency bases, evaporated market depth, and severe cross-venue dislocation. Conversely, fiat USD markets remained highly resilient, proving the shock was an artifact of stablecoin convertibility risk rather than base-asset liquidity. We conclude by evaluating these dynamics against the proposed GENIUS Act, suggesting that while stricter reserve standards can compress stablecoin risk premia, exchange-level frictions and traditional banking bottlenecks will persistently limit market efficiency.

1 Introduction

The March 2023 SVB failure exposed a core weakness in the pre-GENIUS stablecoin regime. Markets treated privately issued dollar tokens as cash-equivalent settlement assets, yet their credibility hinged on opaque reserve management, vulnerable banking counterparties, and narrow redemption channels. When Circle disclosed that a portion of USDC reserves was trapped at SVB, repricing was immediate: USDC broke par, cross-currency wedges opened, and liquidity conditions deteriorated most sharply in stablecoin-quoted markets. This episode offers a clean stress test of how reserve uncertainty propagates into market quality, price alignment, and arbitrage capacity.

This paper studies transmission mechanism using high-frequency BTC spot data across major venues during March 2023. We compare BTC/USD, BTC/USDT, and BTC/USDC markets across calm, stress, and post-stress regimes, focusing on stablecoin premia, cross-currency basis, depth, spread proxies, and cross-venue dislocation. Our central result is that the shock was concentrated in the quote-currency layer rather than in BTC spot trading: fiat BTC/USD markets remained comparatively resilient, while USDC-linked venues experienced the sharpest deterioration in depth, price impact, and market integration.

The episode is also directly relevant for current policy debates. In reduced form, it shows what stablecoin stress looks like in the absence of a fully credible regulatory perimeter: the 1:1 peg breaks,

prices disconnect across stablecoins and exchanges, and arbitrage becomes impaired. This makes the SVB–USDC shock a natural benchmark for evaluating both the intended effects and the limits of the GENIUS Act. The paper outlines the data and empirical framework, presenting evidence on premia, basis, liquidity, and fragmentation, and then drawing out the policy implications.

2 Methodology

The quantitative framework of this research utilizes a high-frequency market microstructure approach to evaluate the dissolution of parity and the resulting liquidity fragmentation during systemic stress events. The study is structured as a *natural laboratory* investigation, centering on the March 2023 Silicon Valley Bank (SVB) collapse which was an exogenous shock exposing the structural vulnerabilities of the pre-GENIUS Act stablecoin ecosystem.

We use high-frequency BTC spot data from Binance and Coinbase over March 1–21, 2023, partitioned into calm (Mar 1–9), stress (Mar 10–14), and post-stress (Mar 15–21) regimes. Raw one-second OHLCV quotes and UTC-aligned tick trades are cleaned of non-positive values and aggregated to one-minute OHLC, volume, log returns, trade counts, VWAP, and signed dollar volume without interpolating empty intervals. Trades are event-time matched to OHLC-derived mid-prices and forward-shifted, providing the basis for all subsequent microstructure measures. We will analyze the data using quantitative measures such as cross-currency basis, depth and spread proxies. We will subsequently perform a fragmentation and a cross-venue dislocation analysis.

2.1 Cross-Currency Basis

We define the **cross-currency basis** in basis points (bps) for exchange e and minute t using one-minute close prices as $\text{basis}_{e,t} = \frac{P_{e,t}^{BTC/USDT} - P_{e,t}^{BTC/USD}}{P_{e,t}^{BTC/USD}} \times 10{,}000$, and focus on its absolute value $|\text{basis}_{e,t}|$, where near-zero values indicate parity and large values capture cross-currency frictions.

We define a transaction cost threshold, $\text{cost}_{e,t}$, which accounts for the cumulative exchange-specific taker fees of a two-leg round-trip arbitrage alongside a regime-dependent slippage and latency buffer. Minutes with $|\text{basis}_{e,t}| > \text{cost}_{e,t}$ are flagged as above-cost basis episodes. For each exchange and regime, we summarize the basis distribution, average cost, percentage of minutes above cost, number and duration of above-cost runs, cross-venue dispersion of the basis, and regression-based sensitivities of $|\text{basis}_{e,t}|$ to regime dummies, BTC/USD inter-venue dislocations, and short-horizon volatility.

2.2 Depth Proxies

We summarize market depth by exchange, pair, and regime using two primary proxies. First, we calculate **Amihud illiquidity** (Amihud, 2002) for each day t as $\text{Amihud}_t = |r_t|/\text{Vol}_t$, where r_t is the daily mid-price return and Vol_t is the corresponding traded notional. To provide an intuitive depth-oriented summary statistic, we report the inverse of the median daily Amihud measure per regime. Second, we estimate a reduced-form **Kyle's** λ to capture price impact through the relation between short-horizon price changes and signed order flow, specified as $\Delta P_t = \alpha + \lambda q_t + \varepsilon_t$. Here, q_t denotes the signed net volume over interval t. Within our one-minute sampling framework, λ

is estimated separately for each cross-section, with larger estimates indicating a greater marginal price impact per unit of order flow.

2.3 Spread Proxies

We characterize trading frictions and market integration across exchange–pair–regimes using five measures. First, the **Roll implied spread** (Roll, 1984) is computed as $\text{Roll} = 2\sqrt{-\widehat{\text{Cov}}(\Delta p_t, \Delta p_{t-1})}$ whenever the first-order auto-covariance of the one-second log mid-price changes (Δp_t) is negative. Second, the **execution deviation proxy** is defined as $ED_i = 2 \cdot side_i \cdot [(P_i - M_i)/M_i] \cdot 10^4$, where P_i is the transaction price, M_i the contemporaneous OHLC-based mid-price, and $side_i \in \{+1, -1\}$ indicates trade direction (trimmed for $ED_i \notin [-500, 500]$ bps). Third, a **realized-spread proxy** for ex post execution cost is constructed using a forward-shifted mid-price $M_{i+\Delta}$: $RS_i^{\text{proxy}}(\Delta) = 2 \cdot side_i \cdot [(P_i - M_{i+\Delta})/M_i] \cdot 10^4$. Fourth, to measure **volume fragmentation**, we calculate the Herfindahl–Hirschman Index $HHI_t = \sum_j s_{j,t}^2$, where $s_{j,t}$ is venue j's volume share. Finally, **cross-venue dislocation** is measured as $disloc_{i,j,t} = |P_{i,t} - P_{j,t}|/[(P_{i,t} + P_{j,t})/2] \cdot 10^4$ using one-minute VWAPs. For pairs traded on three venues, we compute the minute-level mean pairwise dislocation, summarizing the series by regime mean, 90th percentile, and maximum.

3 Empirical Results

This section details the empirical results of our analysis, highlighting how the SVB crisis severely disrupted stablecoin pricing, fragmented market liquidity, and triggered structural dislocations across major cryptocurrency exchanges.

3.1 Stable Coin Dynamics

We examine the pricing behavior of major stablecoins to assess how the banking crisis impacted market confidence and relative valuations. Figure 1 illustrates the USDC de-peg, with the implied premium turning sharply negative during the SVB crisis. While USDT experienced elevated volatility, it maintained a positive premium. The persistent USDT premium gap between Binance and Coinbase suggests ongoing cross-exchange arbitrage, driving a spike in the implied USDT/USDC cross-rate premium that reflects a sudden shift in relative stablecoin confidence.

Pre-crisis, USDC traded at a near-perfect peg (+0.2 bps). During the crisis, it crashed to an average of -197.0 bps, with extreme negative deviations stretching to nearly -1,250 bps. Post-crisis, USDC recovered to -3.9 bps, suggesting lingering uncertainty despite Circle's confirmation of full redemption. Conversely, USDT premiums surged to +46.2 bps on Binance and +43.9 bps on Coinbase during the stress regime, driven by a flight-to-safety from USDC and market-maker liquidity constraints.

Notably, USDT remained elevated post-crisis (+27.5 bps on Binance, +21.3 bps on Coinbase). Table 1 highlights the severity of this dislocation. The implied USDT/USDC cross-rate averaged +256.7 bps during the crisis, peaking at over 1,500 bps (an explicit 15-cent premium for USDT in BTC-implied terms). The failure of these premiums to revert to the calm pre-crisis baseline demonstrates significant hysteresis in stablecoin confidence. The market did not fully re-equalize,

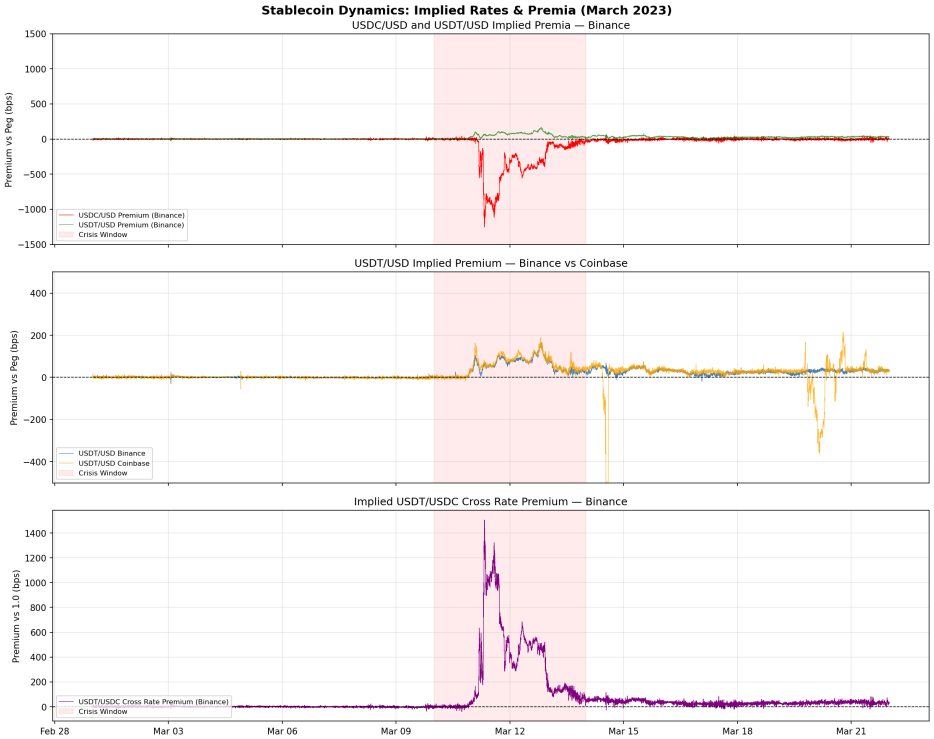


Figure 1: Implied Rates and Premia (March 2023)

traders retained a structural preference for USDT, resulting in a durable repricing of relative stablecoin risk.

3.2 Cross Currency Basis

We examine the cross-currency basis between stablecoin and fiat Bitcoin markets to assess market efficiency and the persistence of arbitrage opportunities. Figure 3 plots the 60-minute rolling BTC/USDT–BTC/USD basis across Binance, Coinbase, and Kraken. Figure 4 illustrates the share of one-minute observations exceeding an exchange- and regime-specific transaction cost threshold. We modeled a two-leg, round-trip basis trade requiring four taker fills. Assuming taker fees are 1 bp (Binance), 25 bps (Coinbase), and 22 bps (Kraken), plus an execution-friction buffer of 12 bps for calm/post-crisis and 25 bps for the stress regime.

Before costs, all venues traded near parity in the calm regime (-0.1 to -0.4 bps). During stress, the mean basis widened to between -45 and -54 bps and remained materially negative post-crisis. Tail risk increased concurrently, with the 90th percentile absolute basis exceeding 90 bps across all three venues. This reflected a cross-currency wedge that only partially normalized post-crisis.

Adjusting for trading costs, the persistence of actionable arbitrage is highly venue-dependent. On Binance, the low fee structure allowed the absolute basis to clear the cost threshold in 63% of stress-regime minutes and 87% post-crisis, yielding a positive mean net edge (+16.7 and $+11.5bps$, respectively). Conversely, the higher fees on Coinbase and Kraken restricted profitable opportunities to just 3–5% of the stress period, with average net edges remaining negative across all regimes. Thus, while gross stress was market-wide, economic persistence was concentrated in

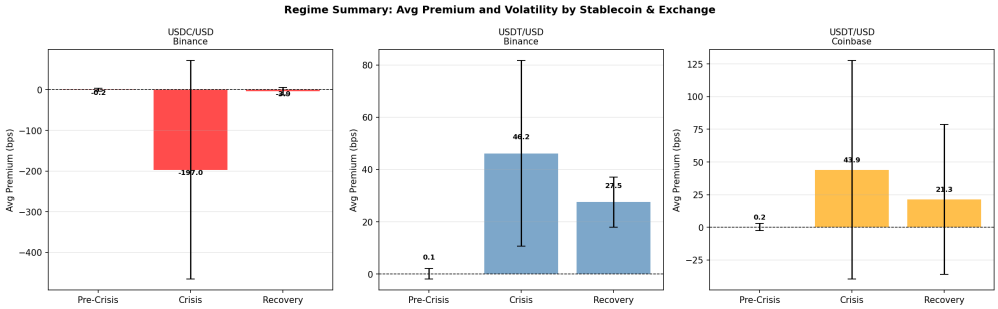


Figure 2: Regime summary across Exchanges

Table 1: Implied Stablecoin Premium Summary by Exchange, Instrument, and Regime

Exchange	Instrument	Regime	Avg (bps)	Std (bps)	Min (bps)	Max (bps)
Binance	USDC/USD	Pre-Crisis	−0.2	3.7	−32.3	34.1
		Crisis	−197.0	268.7	−1,251.2	65.3
		Recovery	−3.9	9.3	−61.6	50.2
Binance	USDT/USD	Pre-Crisis	0.1	2.1	−30.6	25.1
		Crisis	46.2	35.5	−9.5	165.9
		Recovery	27.5	9.6	−18.5	75.6
Coinbase	USDT/USD	Pre-Crisis	0.2	2.7	−56.2	30.0
		Crisis	43.9	83.6	−742.1	187.7
		Recovery	21.3	57.4	−361.2	214.8
Binance	USDT/USDC	Pre-Crisis	0.3	3.8	−36.0	30.9
		Crisis	256.7	315.0	−38.8	1,504.5
		Recovery	31.5	12.2	−20.4	88.2

Notes: Implied stablecoin premiums are computed as $(P_t^{BTC/USD}/P_t^{BTC/X} - 1) \times 10{,}000$ bps, where $X \in \{\text{USDC}, \text{USDT}\}$. A positive value indicates the stablecoin traded above its \$1.00 peg; a negative value indicates a discount. Regimes are: Pre-Crisis (Mar 1–9), Crisis (Mar 10–14), Recovery (Mar 15–21). Data are sampled at one-minute frequency. N = 12,960 (Pre-Crisis), 7,200 (Crisis), 10,080 (Recovery) observations per series.

low-fee venues.

Episode duration statistics reinforce this. Binance experienced hundreds of above-cost basis episodes during and after the crisis, averaging 30–33 minutes, with extreme cases exceeding 1,000 minutes. Coinbase and Kraken see only brief, stress-only bursts lasting under 10 minutes. This demonstrates that in low-fee environments, cross-currency wedges can persist for extended periods, whereas high-fee venues either fail to clear cost thresholds or see deviations quickly corrected.

3.3 Market Depth and Spread

An analysis of order-book absorption and price impact reveals that fiat pairs exhibited substantially greater liquidity depth than their stablecoin counterparts. As illustrated in Figure 5 (Median Inverse Amihud Illiquidity), fiat USD serves as the dominant liquidity anchor: the inverse Amihud depth for Coinbase BTC/USD exceeds all other pairs by several orders of magnitude, a divergence clear

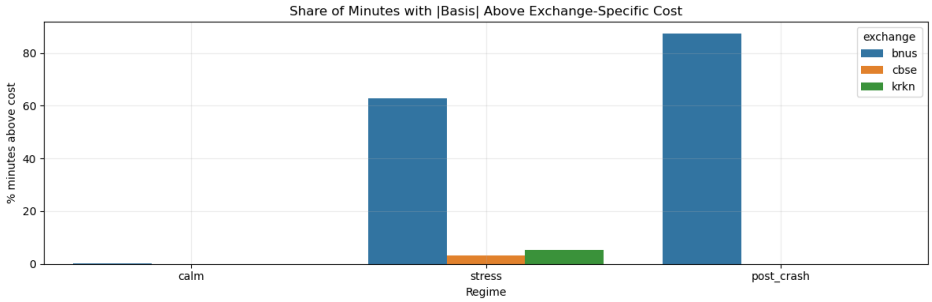


Figure 3: Basis time series: BTC/USDT vs BTC/USD Basis Over Time (60-min rolling mean)

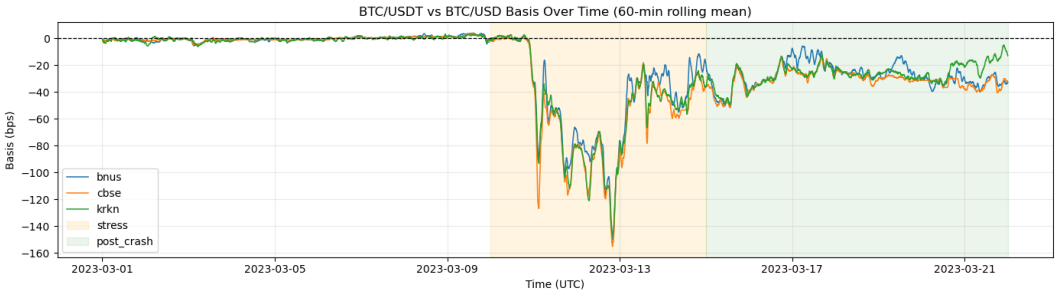


Figure 4: Share of Minutes with $|Basis|$ Above Exchange-Specific Cost.

even on a logarithmic scale.

Furthermore, fiat pairs demonstrated structural resilience during the stress episode. At the peak of the banking turmoil, the Amihud median for Coinbase BTC/USD improved marginally from 1.87×10^{-9} to 1.52×10^{-9}, indicating the fiat order book absorbed the volume surge seamlessly. Conversely, stablecoin-quoted pairs lacked this robustness. As the USDC peg and its banking counterparties faced pressure, stablecoin depth metrics deteriorated as a result.

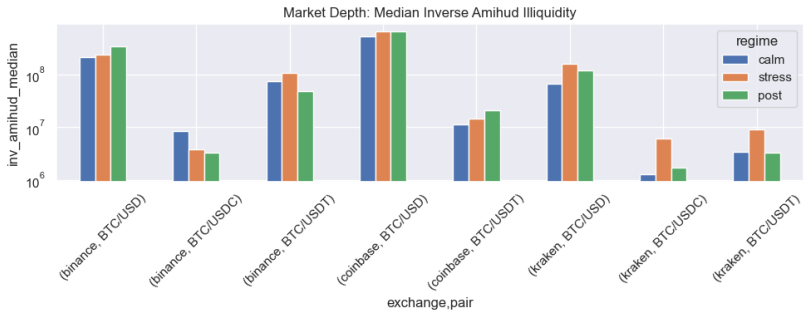


Figure 5: Median Inverse Amihud

Spread proxies further underscore severe liquidity fragmentation across quote currencies. Figure 7 illustrates the evaporation of stablecoin liquidity during the crisis. The Binance BTC/USDC Roll implied spread surged from a calm-regime baseline of 0.86 bps to 17.2 bps during the stress window, an approximate 20-fold expansion in implied trading frictions. In stark contrast, Coinbase BTC/USD maintained the tightest spreads across all regimes, exhibiting only modest widening. Execution friction was thus entirely isolated to the stablecoin quote currency.

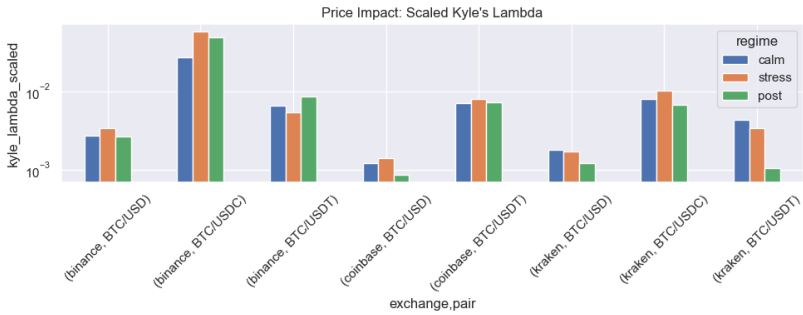


Figure 6: Kyle's Lambda

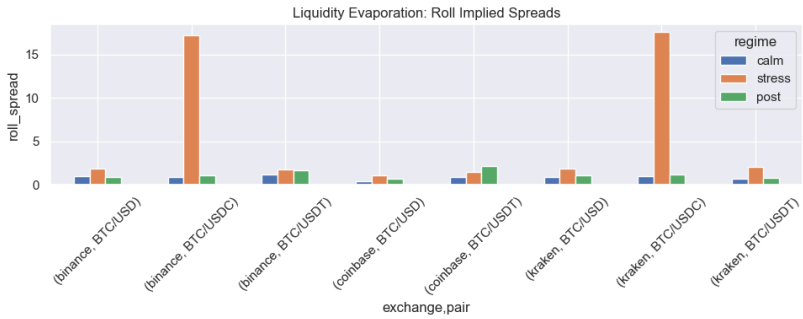


Figure 7: Liquidity Evaporation

While signed execution-deviation and realized-spread proxies remained near zero under standard conditions, they exhibited significant instability during the SVB stress window, particularly within USDC-quoted markets (Figure 8). Because these measures rely on one-second OHLC-derived mid-price proxies rather than Level-2 best-bid/best-offer (BBO) quotes, they are subject to intra-second latency artifacts. Consequently, they serve as supplementary diagnostics for adverse selection rather than literal transaction costs. The Roll implied spread, Kyle's λ, and Amihud illiquidity remain the primary, robust measures of market friction.

Across all liquidity proxies, the largest stress-time deterioration is concentrated in USDC-quoted markets (notably Binance BTC/USDC), confirming that the shock propagated primarily through the quote-currency layer rather than base-asset spot risk.

3.4 Volume Fragmentation

We examine the fragmentation of trading volume across different quote currencies to assess the impact the crisis had on the liquidity market. Figure 9 plots the 6-hour rolling venue volume shares, demonstrating that crisis-induced fragmentation was concentrated almost entirely within the USDC market. For BTC/USD, Coinbase maintained a commanding market share across calm (60%), stress (58%), and post-crisis (55%) regimes, with shifts falling below our 5-percentage-point significance threshold. Similarly, Binance dominates BTC/USDT throughout, declining only gradually from 75.2% (calm) to 70.4% (stress). Neither fiat nor USDT pairs experience a stress-time leadership flip, exhibiting only mild post-crisis deconcentration.

Conversely, the BTC/USDC market undergoes a severe structural migration. While Binance

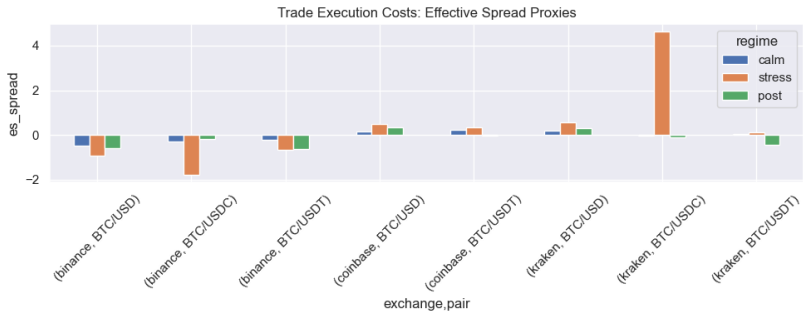


Figure 8: Trade Execution Cost

dominates the calm regime (≈ 67.8%), volume shares sharply invert during the stress window as Kraken captures the majority share (≈ 58.2%). Coupled with a Herfindahl-Hirschman Index (HHI) spike from 0.63 to 0.79, this confirms that USDC liquidity not only migrated but became highly concentrated amidst the banking panic.

3.5 Cross Venue Dislocation

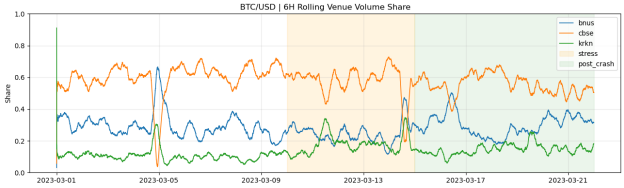
At the one-minute VWAP level, mean pairwise cross-venue dislocation in BTC/USD increases from 1.6 bps in the calm regime to 7.8 bps under stress. This approximate five-fold widening improves only partially to 5.8 bps post-crisis, indicating a persistent integration break even after macro conditions stabilized. BTC/USDT exhibited a similar but muted pattern. Mean dislocation rose from 1.7 bps to 3.8 bps during stress, settling at 3.3 bps post-crisis to reflect moderate, durable fragmentation.

The most severe breakdown occurs in BTC/USDC where the mean Binance–Kraken dislocation surged from 3.5 bps to over 52 bps during the panic—an order-of-magnitude shock. Despite falling to 6 bps post-crisis, USDC cross-venue misalignment remains structurally elevated relative to both fiat and USDT pairs.

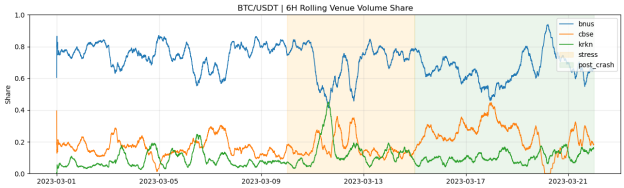
As illustrated in Figure 10, this stress-time integration failure is concentrated in the USDC market. These findings reinforce the conclusion that the SVB episode manifested primarily as a structural price-alignment breakdown across venues, rather than a broad-based collapse in base-asset trading activity.

4 Limitations of the Regulatory Framework

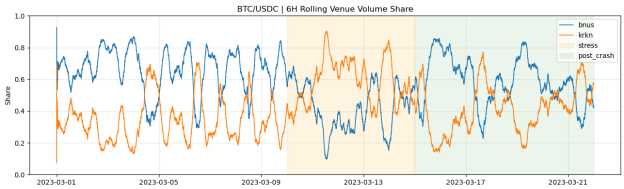
While the GENIUS Act materially improves transparency and baseline solvency, it would not have fully insulated stablecoins from the March 2023 microstructure breakdown. The $USDC$ de-peg was fundamentally a *commercial-bank counterparty event*: without direct access to Federal Reserve master accounts, 1:1 reserve requirements simply relocate risk from on-chain balances to uninsured deposits at fractional-reserve banks, leaving payment stablecoins exposed to traditional bank-run dynamics. In addition, regulation cannot bridge the *temporal gap* between 24/7 crypto markets and limited-hours fiat rails. The extreme widening in $BTC/USDC$ Roll spreads was amplified by the weekend closure of systems such as Fedwire, which prevented arbitrageurs from minting and



(a) BTC/USD



(b) BTC/USDT



(c) BTC/USDC

Figure 9: Volume Fragmentation Across Quote Currencies

redeeming to enforce the peg; similar fragmentation will likely persist in future banking panics until round-the-clock fiat settlement is available alongside the GENIUS framework.

There are also *adoption and international alignment* challenges. The benefits of a safer stablecoin perimeter arise only if major exchanges, market makers, and payment networks migrate to GENIUS-compliant tokens; continued reliance on unregulated or offshore stablecoins would leave large parts of the market subject to the same convertibility frictions seen in March 2023. Internationally, the Act effectively privileges dollar-pegged payment stablecoins, while other jurisdictions—most notably the EU under MiCA and potential digital-euro arrangements—are building their own regimes for euro- and multi-currency stablecoins (International Law, 2025). If foreign regulators are reluctant to anchor their infrastructures to a U.S.-centric standard, global liquidity may fragment into parallel USD- and non-USD-stablecoin zones, so that cross-currency wedges similar to those in our data reappear at the boundaries between regulatory regimes even if they shrink within the GENIUS perimeter.

5 Conclusion and Policy Implications

Our empirical findings demonstrate that, in the absence of federal regulation, stablecoins acted as vectors for traditional financial contagion, severely disrupting digital asset market microstructure. During the March 2023 banking stress, uncertainty surrounding USDC's reserve backing triggered a massive liquidity flight, causing stablecoin volatility to surge to 200% and bid-ask spreads

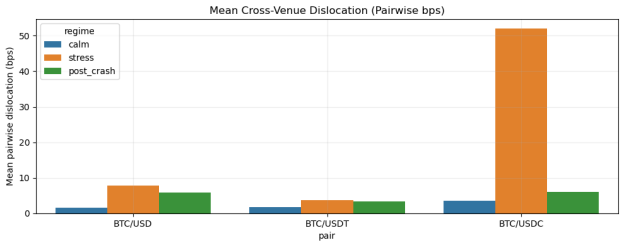


Figure 10: Cross Venue Dislocation

to widen by over 20-fold. Crucially, fiat USD order books maintained robust depth and stable execution costs, proving the disruption was driven by stablecoin-specific credit risk rather than base-asset valuation. The primary distortions were convertibility and pricing frictions. USDC pairs exhibited severe deterioration across Amihud depth, Kyle's λ, and cross-venue dislocation, while the BTC/USDT–BTC/USD basis widened into large, persistent wedges.

By mandating strict reserve backing, transparent auditing, and direct oversight, the GENIUS Act targets the informational asymmetry and default risk that prompted market makers to pull liquidity during the SVB crisis. Under this framework, stablecoins shift from speculative credit instruments to reliable settlement rails, so the stablecoin-specific risk premium should compress. With reserve quality assured, cross-currency arbitrage between BTC/stablecoin and BTC/USD depends less on ad-hoc counterparty confidence and more on standard microstructure frictions such as fees, latency, and inventory which should yield smaller, shorter-lived pricing wedges and more symmetric information flow across venues.

Yet our cost-adjusted basis results underscore the persistence of venue-specific frictions. Even with fully regulated stablecoins, arbitrage profitability diverges sharply between low-fee venues (Binance) and higher-cost venues (Coinbase, Kraken) because of structural fee tiers and fiat on/off ramp bottlenecks. Thus, while the GENIUS framework can substantially reduce the credit and opacity components of market dislocation, it cannot eliminate exchange-level microstructure constraints. A realistic policy expectation is a marked attenuation of stress-time pricing wedges, not the disappearance of cross-venue and cross-currency price dispersion.

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